Hamiltonian reduction

§ Recall

Poisson variety/mfd

$G \curvearrowright X \Rightarrow \mathfrak{g} \longrightarrow \mathrm{Vect}(X)$

$\mu^* \dashrightarrow \mathcal{F}(X)$, $\mathcal{F}(X) \to \mathrm{Vect}(X)$ via φ

Moment map $\mu : X \longrightarrow \mathfrak{g}^*$ is s.t. $\varphi(\langle \mu(x), \xi \rangle) = \mathbb{R}_{\mathfrak{X}} \mathfrak{X}_\xi(x)$

- Prop.

$$\begin{array}{ccc} X & \xrightarrow{g} & X \\ \downarrow \mu & \circlearrowright & \downarrow \mu \\ \mathfrak{g}^* & \xrightarrow[Ad_g^*]{} & \mathfrak{g}^* \end{array}$$

Example

Recall symplectic structure on T^*X.

Given an action $G \curvearrowright X$ it naturally extends to the one on T^*X so that:

- $\pi : T^*X \longrightarrow X$ is G-equivariant
- $G \curvearrowright X$ preserves the one-form $\Theta = \sum_{i=1}^{n} p_i dx_i$.

Exercise Check it.

Prop This induced action is Hamiltonian w/ moment map

$\mu : \langle \mu(\alpha), \xi \rangle = \langle \alpha, V_\xi(\pi(\alpha)) \rangle$, where $V_\xi(x)$ is the image of $\xi \longrightarrow V_\xi(x)$

Proof. Note that $\langle \alpha, V_\xi(\pi(\alpha))\rangle = \langle \theta, \tilde{V}_\xi(\alpha)\rangle$ ← Lift of V_ξ into $T_\alpha T^*X$.

$$d(\langle \mu, \xi\rangle) = -d\, 2_{\tilde{V}_\xi}\theta = -\underbrace{(d2_{\tilde{V}_\xi} + 2_{\tilde{V}_\xi} d)(\theta)}_{\mathcal{L}_{\tilde{V}_\xi}(\theta)=0} + 2_{\tilde{V}_\xi} d\theta =$$

$$= 2_{\tilde{V}_\xi}\omega = \omega(\tilde{V}_\xi, \cdot)$$ ∎

Examples.

(1) $G = \mathbb{C}^\times \curvearrowright \mathbb{C}^n$: $\lambda : (x_1, \dots, x_n) \longmapsto (\lambda x_1, \dots, \lambda x_n)$

$\mathfrak{g} = \mathbb{C} = \langle 1\rangle \cong \mathfrak{g}^*$ $\Rightarrow V_\xi(x) = \frac{d}{dt}\big|_{t=0}\left((e^{t\xi}x_1, \dots, e^{t\xi}x_n)\right) = \xi\cdot(x_1, \dots, x_n)$

$\Rightarrow G \curvearrowright T^*\mathbb{C}^n : (p_1, \dots, p_n, x_1, \dots, x_n) \longmapsto (\lambda^{-1}p_1, \dots, \lambda^{-1}p_n, \lambda x_1, \dots, \lambda x_n)$.

$$\langle \mu(p,x), \xi\rangle = -\langle \sum_{i=1}^n p_i dx_i, \xi\cdot(x_1, \dots, x_n)\rangle = \left(-\sum_{i=1}^n p_i x_i\right)\cdot\xi.$$

$\mu(p,x) \in \mathbb{C}$

(2) Exercise Compute the moment map for the action of $G = \mathbb{C}^n$ induced on $T^*\mathbb{C}^n$ from $G \curvearrowright \mathbb{C}^n$ by translations.

§ Hamiltonian reduction. (Hamiltonian, Poisson.)

Motivation: We have $G \curvearrowright X$.

Assume $H \in \mathcal{F}(X)^G$. $\Rightarrow$ G maps integral curves of H into integral curves of H. Let V_H be the v.f. corr. to H.

Want to study induced dynamics on "X/G".

- Let $\mathcal{O} \subset \mathfrak{g}^*$ be an orbit.

Then $\mu^{-1}(\mathcal{O})$ is G-invariant.

- Level-sets of μ are V_H-invariant.

Problem: $\mu^{-1}(\mathcal{O}) \subset X$ in general is not Poisson/symplectic.

However $\mu^{-1}(\mathcal{O})/G$ often is.

Def. Assume $\mu: X \to \mathfrak{g}^*$ is a moment map.
$\mathcal{O} \subset \mathfrak{g}^*$ is a closed orbit.
Then the hamiltonian reduction is

$$X /\!\!/_{\mathcal{O}} G := \mu^{-1}(\mathcal{O})/G.$$

Algebraically we study $F(X)^G / (F(X)^G \cap \ldots)$ $\left(\text{or rather } \left(F(X)/\mathbb{I}(\mathcal{O})\right)^G\right)$ ($\mathbb{I}(\mathcal{O}) = \mathbb{I}(\mu^{-1}(\mathcal{O}))$, $\mu^*(\mathbb{I}(\mathcal{O}))$)

Or rather $F(X)^G / (F(X)^G \cap \mu^*(\mathbb{I}(\mathcal{O})))$.

Prop. $F(X)^G \cap (\mu^*(\mathbb{I}(\mathcal{O}))) \trianglelefteq F(X)^G$ is a Poisson ideal. (Exercise)
$\left(F(X)/(\mu^*(\mathbb{I}(\mathcal{O})))\right)^G$ has a natural structure of Poisson algebra.

Proof. • Note that $(\mu^*(\mathbb{I}(\mathcal{O})))$ is G-invariant: $g^*(\sum_{i=1}^n a_i b_i \circ \mu) = \sum_{i=1}^n g^*(a_i) \cdot (\mathrm{Ad}_g^*)^*(b_i) \circ \mu$. ($\mathbb{I}(\mathcal{O})$)

Let $f \in F(X)$ be s.t. $[f]$ is G-invariant $\Leftrightarrow [(e^{t\xi})^*(f)] = [f] \ \forall \xi \in \mathfrak{g}, t \in \mathbb{R}$,
$\Leftrightarrow \frac{d}{dt}|_{t=0} ([(e^{t\xi})^*(f)]) = 0$
$\Leftrightarrow [v_\xi(f)] = 0$ ($\Leftarrow$)
($\Leftarrow$) $\{\langle \mu, \xi \rangle, f\} \in (\mu^*(\mathbb{I}(\mathcal{O}))) \ \forall \xi \in \mathfrak{g}$.

Then set

$$\{[f],[h]\} = [\{f,h\}].$$

Note $\left[\{f + \sum_{i=1}^n a_i (b_i \circ \mu), h\}\right] = [\{f,h\}] + \left[\sum_{i=1}^n \{a_i, h\} \underbrace{b_i \circ \mu}_{\in \mu^*(\mathbb{I}(\mathcal{O}))} + \right.$
$\left. + \sum_{i=1}^n a_i \underbrace{\{b_i \circ \mu, h\}}_{= \sum_k \frac{\partial b_i}{\partial y_k} \cdot \{\mu_k, h\} \in (\mu^*(\mathbb{I}(\mathcal{O})))} \right] = [\{f,h\}]$.

$b_i \in \mathbb{I}(\mathcal{O})$

∎

Theorem

Let X be symplectic, $G \curvearrowright X$ Hamiltonian action free on $\mu^{-1}(0)$ where $0 \in \mathfrak{g}^*$ closed orbit. Let $H \in \mathcal{F}(X)^G$

Then

- $X /\!\!/_0 G$ is symplectic w/ a form $\tilde{\omega}$ s.t. $\pi^*(\tilde{\omega}) = \omega|_{\mu^{-1}(0)}$ where $\pi : \mu^{-1}(0) \longrightarrow \mu^{-1}(0)/G$.
- The dynamics induced by hamiltonian H descends to the dynamics on $X /\!\!/_0 G$ induced by $\tilde{H}$ s.t. $\pi^*(\tilde{H}) = H|_{\mu^{-1}(0)}$.

∎

Prop. Assume $G \curvearrowright X$ (smooth mfld/vty, affine) freely.

Then $T^*X /\!\!/_0 G \cong T^*(X/G)$.

Proof [sketch].

Consider the case $X = Y \times G$, with $G \curvearrowright Y \times G$ on the second component by the left shifts.

Then $T^*X = T^*Y \times T^*G$, $\langle \mu(\lambda,\beta), \xi \rangle = -\langle \beta, v_\xi \rangle \Rightarrow$

λ ↓ (λ,β)

$\Rightarrow \mu^{-1}(0) = \{(\lambda,\beta) \mid \langle \beta, v_\xi \rangle = 0 \ \forall \xi \in \mathfrak{g}\} \cong G \in T^*G.$

Note $T^*G \cong G \times \mathfrak{g}^* \Rightarrow \mu^{-1}(0) = T^*Y \times (G \times \{0\})$

$\Rightarrow \mu^{-1}(0)/G = T^*Y$. But $Y = X/G$. ∎

rop.

Let $G \curvearrowright X$ - Hamiltonian action. Let $\mathcal{O} \subset \mathfrak{g}^*$ be a closed orbit.

Then $X /\!\!/_{\mathcal{O}} G \cong (X \times (-\mathcal{O})) /\!\!/_{\mathcal{O}} G$, where $g: (x, \alpha) \longmapsto (g.x, Ad^*_g(\alpha))$. ($X$, $-\mathcal{O}$)

Proof.

- $\mu_{X \times (-\mathcal{O})} = \mu_X + \mu_{-\mathcal{O}}$ ↑
- $\mu_{-\mathcal{O}} = Id_{-\mathcal{O}}$ ← exercise
- $\mu_{X \times (-\mathcal{O})}^{-1}(0) = \{(x, \alpha) \mid \mu_X(x) + \mu_{-\mathcal{O}}(\alpha) = 0\} \xrightarrow[P_1]{\cong} \{x \in X \mid \mu_X(x) \in \mathcal{O}\}$ (α = $\mu_{-\mathcal{O}}(\alpha)$; $\cong$ ↓ G-equivariant.)

$= \mu^{-1}(\mathcal{O}) \Rightarrow \mu_{X \times (-\mathcal{O})}^{-1}(0)/G \cong \mu^{-1}(\mathcal{O})/G$. ∎

Corollary Assyme that $G \curvearrowright X$ - $\mathcal{O} \cong T^*N$, as a G-space. and G acts on $X \times N$ freely.

Then $T^*X /\!\!/_{\mathcal{O}} G \cong T^*((X \times N)/G)$

Proof. $T^*X /\!\!/_{\mathcal{O}} G \cong (T^*X \times (-\mathcal{O})) /\!\!/_{\mathcal{O}} G \cong (T^*X \times T^*N) /\!\!/_{\mathcal{O}} G \cong T^*((X \times N)/G) \cong$

$\cong T^*((X \times N)/G)$. ∎.

Example

Consider the action of $G \curvearrowright G$ by the left shifts.

It induces $G \curvearrowright T^*G \cong G \times \mathfrak{g}^*$ - Hamiltonian action.

Assume that the fibers are identified w/ right shifts.

Then $g: \underset{\in G}{(h}, \underset{\in \mathfrak{g}^*}{\lambda}) \longmapsto (g.h, Ad_g^*(\lambda))$.

The moment map $\mu: (h, \lambda) \longmapsto \lambda \Rightarrow$

$\Rightarrow \mu^{-1}(\mathcal{O}) \cong G \times \mathcal{O}$.; $\mu^{-1}(\mathcal{O})/G \cong \mathcal{O}$.

Cor. $\mathcal{O}$ is symplectic.

Exercise Compute the symplectic form on $\mathcal{O}$.

Example

Recall the action $\mathbb{C}^\times \curvearrowright \mathbb{C}^n$. , $\mathcal{O} = \{1\} = T^*(pt)$

Consider $T^*\mathbb{C}^n /\!\!/_{-1} \mathbb{C}^\times = T^*\left((\mathbb{C}^n \times \{1\})/\mathbb{C}^\times\right) = T^*(\mathbb{P}^{n-1})$.

Also: $T^*\mathbb{C}^n /\!\!/_{-1} \mathbb{C}^\times = \{(p,x) \mid \langle p,x\rangle = 1\}/\mathbb{C}^\times \cong \{p \otimes x \mid Tr(p \otimes x) = 1\}$ -

- (Co)adjoint orbit in $\mathfrak{gl}_n^*$. Denote this orbit $\mathcal{O}_n$.

Example

Consider $G = \mathbb{P}GL_n \curvearrowright T^*\mathfrak{gl}_n$ and

$$T^*\mathfrak{gl}_n /\!\!/_{\mathcal{O}_n} G \cong T^*\left(\mathfrak{gl}_n \times \mathbb{P}^{n-1}/G\right)'' \cong T^*(\mathfrak{h}_{reg}) = T^*(\mathbb{C}^n \setminus \Delta)$$

$T^*\mathfrak{gl}_n \cong \mathfrak{gl}_n^{\times 2} = \{(p,x)\}$.

Consider $H_i = Tr(p^i) \in \mathcal{F}(T^*\mathfrak{gl}_n)^G \Rightarrow \widetilde{H}_i$ - integrable system (Calogero-Moser).

§ Quantum Hamiltonian reduction.

Let H be a Hopf algebra, A-assoc. algebra.

Def. Moment map is a homomorphism

$$\mu : H \longrightarrow A.$$

- A is an H-module through "adjoint action".

$$h : a \longmapsto \mu(h_{(1)}) \cdot a \cdot \mu(S(h_{(2)})).$$

Example Assume $G \curvearrowright X$ (mfd.) $\Rightarrow \mathfrak{g} \longrightarrow \mathrm{Vect}(X)$.

This induces $\mu : \mathcal{U}(\mathfrak{g}) \longrightarrow D(X)$.

- $D(X)$ is a "quantization" of F $(T^*(X))$.

Def. Let $\mu : H \longrightarrow A$ be a moment map.

Let $I \subset H$ be a two-sided ideal.

$$A /\!\!/_I H := \left(A / A \cdot \mu(I) \right)^H.$$

- $A \cdot \mu(I)$ is invariant w/ adjoint action of H:

$$\mu(h_{(1)}) a \cdot \mu(b) \mu(S(h_{(2)})) = \underbrace{\mu(h_{(1)}) \cdot a}_{\in A} \cdot \mu(\underbrace{b}_{\in I} \cdot S(h_{(2)})).$$

- Prop. $A /\!\!/_I H$ inherits a natural structure of an associative algebra.

Proof. Assume $[a] \in \left(A / A\mu(I) \right)^H$. Then $\mu(h_{(1)}) a \mu(S(h_{(2)})) \in \epsilon(h) a + A \cdot \mu(I)$. Want: i) $\mu(b) \cdot a \in A \cdot \mu(I) \ \forall b \in I$.

$$\mu(b) \cdot a = \mu(b_{(1)}) \epsilon(b_{(2)}) \cdot a = \mu(b_{(1)}) \cdot a \cdot \epsilon(b_{(2)}) = \mu(b_{(1)}) \cdot a \cdot \mu(S(b_{(2)(1)})) \cdot \mu(b_{(2)(2)}) = \mu(b_{(1)(1)}) \cdot a \cdot \mu(S(b_{(1)(2)})) \cdot \mu(b_{(2)}) \in \epsilon(b_{(1)}) \cdot a \cdot \mu(b_{(2)}) + A \cdot \mu(I) =$$

$$= a \cdot \epsilon\mu(b) + A\mu(I) = A \cdot \mu(I).$$

Then $(a_1 + A\cdot\mu(I))(a_2 + A\cdot\mu(I)) = a_1\cdot a_2 + A\cdot\mu(I)$.

Need to check: $[a_1][a_2] \in \left(A/A\mu(I)\right)^H \Rightarrow [a_1\cdot a_2] \in \left(A/A\mu(I)\right)^H$.

Indeed: $\mu(h_{(1)})\cdot a_1\cdot a_2\cdot\mu(S(h_{(2)})) = \mu(h_{(1)_{(1)}})\, a_1\, \epsilon(h_{(1)_{(2)}})\cdot a_2\mu(S(h_{(2)}))=$

$= \mu(h_{(1)_{(1)}})a_1\,\mu(S(h_{(1)_{(2)_{(1)}}}))\,\mu(h_{(1)_{(2)_{(2)}}})\,a_2\mu(S(h_{(2)})) =$

$= \mu(h_{(1)_{(1)_{(1)}}})a_1\,\mu(S(h_{(1)_{(1)_{(2)}}}))\,\mu(h_{(1)_{(2)}})\,a_2\mu(S(h_{(2)})) \in$

$\in a_1\,\epsilon(h_{(1)_{(1)}})\,\mu(h_{(1)_{(2)}})\,a_2\mu(S(h_{(2)})) + \underbrace{A\cdot\mu(I)\cdot a_2\cdot\mu(S(h_{(2)}))}_{\subseteq A\cdot\mu(I)} =$

$= a_1\cdot\mu(h_{(1)})\,a_2\mu(S(h_{(2)})) + A\cdot\mu(I) = \epsilon(h)a_1\cdot a_2 + A\cdot\mu(I)$. ∎

S-invariant

Prop. Let $I \subset H$ - 2-sided ideal, $\mu: H \to A$ - a moment map.
Let V_I be an H/I module, V - an A-module.
Then $(V_I \otimes V)^H$ is an $A/\!/_I H$-module.

Proof.
Define the action by $[a].v = (1\otimes a).v$
Check: $\forall b \in I$, $(1\otimes\mu(b)).v = (1\otimes\epsilon(b_{(1)})\mu(b_{(2)})).v =$

$= (S(b_{(1)_{(1)}})\,b_{(1)_{(2)}}\otimes\mu(b_{(2)})).v = (S(b_{(1)})\,b_{(2)_{(1)}}\otimes\mu(b_{(2)_{(2)}})).v =$

$= (S(b_{(1)})\epsilon(b_{(2)})\otimes 1).v = (S(b)\otimes 1).v = 0$. ∎